The Psychology of Automated Trust in Life-Critical Sport:

Fear of Falling, Algorithm Aversion, and the

Absent Lead Auto-Belay

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Abstract (Human Factors version)

Abstract

Indoor lead climbing presents a distinctive convergence of physical risk, social dependency, and technological absence that has not been adequately addressed in the human factors literature. Unlike top-rope climbing, lead climbing requires a human belayer at all times: no commercially available, U.S.-certified automated substitute exists despite documented and growing market demand. We argue that this absence reflects, in part, a psychologically maintained equilibrium sustained by three interlocking factors: opacity aversion — the well-documented resistance people extend to life-critical systems whose internal mechanisms they cannot observe; fear of falling as a pre-cognitive state that is not addressable by rational reassurance and whose reduction requires progressive fall exposure that existing infrastructure impedes; and the belay partnership as a social structure whose removal triggers trust deficits extending beyond the mechanical catching function. Drawing on the Technology Acceptance Model (Davis, 1989; Venkatesh & Davis, 2000), the automation trust and algorithm aversion literatures (Lee & See, 2004; Parasuraman & Riley, 1997; Dietvorst et al., 2015; Dzindolet et al., 2002), and physiological fear-of-falling research (Pijpers et al., 2003; Nieuwenhuys et al., 2008; Draper et al., 2010; Hodgson et al., 2009), we introduce ARIA (Autonomous Rope Intelligence Architecture) as a design case study in engineering a life-critical system whose architecture directly responds to identified psychological constraints. We further apply Mayer et al.'s (1995) trust model to map ARIA's design features onto the ability, benevolence, and integrity dimensions of perceived trustworthiness. Implications for safety-critical automation design, structured onboarding protocols, and climbing pedagogy are discussed.

Keywords: lead climbing, fear of falling, algorithm aversion, automation trust, opacity aversion, technology acceptance, auto-belay, human factors, safety-critical systems, trust calibration

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Abstract

Fear of falling (FoF) in indoor lead climbing is a well-documented performance-limiting phenomenon that operates at a pre-cognitive level, disrupting visual attention, movement efficiency, and decision-making independent of objective fall risk (Pijpers et al., 2003, 2005; Nieuwenhuys et al., 2008). Progressive fall exposure is the most effective intervention for FoF reduction, yet partner dependency, social inhibition, and gym infrastructure systematically restrict access to the high-frequency fall practice that exposure requires. This paper characterizes the structural barriers to fall practice in indoor lead climbing, with particular attention to a technological gap that reinforces those barriers: no U.S.-certified lead auto-belay device exists, leaving the one infrastructure change most likely to increase fall-practice accessibility unrealized. We argue that this absence is maintained not only by engineering and regulatory challenges but by the same psychological forces that make FoF itself difficult to address: climbers' well-documented distrust of opaque automated systems in life-critical contexts, and the social and psychological functions of the human belay partnership that an automated replacement must respect. Physiological evidence comparing lead and top-rope climbing (Draper et al., 2010; Hodgson et al., 2009) and fall-training exposure data (Niedermeier et al., 2019) establish the urgency of the access problem. ARIA (Autonomous Rope Intelligence Architecture), a hybrid mechanical–electronic lead auto-belay device, is introduced as a design case study whose features were explicitly shaped by the sport psychology literature. Implications for FoF intervention design, climbing pedagogy, and sport-technology adoption are discussed.

Keywords: fear of falling, lead climbing, fall practice, exposure therapy, auto-belay, sport psychology, technology acceptance, algorithm aversion, belay partnership, psychophysiology

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1 Introduction

There is a paradox at the center of indoor lead climbing. The sport is practiced by millions of people in hundreds of gyms purpose-built to support it. Yet the single piece of technology that would make it accessible to solo visitors — an automated device to catch their falls — does not commercially exist in the United States. Every other major gym activity has been decoupled from partner dependency: treadmills replaced running partners, weight machines replaced spotters, and top-rope auto-belays let climbers ascend the wall alone. Lead climbing — the most technically advanced and progression-relevant discipline available on an indoor wall — has remained stubbornly dependent on another human being.

This paper asks why, and the answer is not primarily mechanical.

The Climbing Wall Association identified 577 commercial climbing facilities in the United States and Canada as of 2019, growing at nearly 9% per year (Climbing Wall Association, 2021), with the Outdoor Foundation estimating over 5.6 million Americans identifying as indoor climbers as of 2023 (Climbing Wall Association, 2023). Of those facilities, a substantial proportion have dedicated lead walls that, by the nature of the discipline, can only be used when two compatible partners arrive simultaneously. None can today purchase a certified lead auto-belay device for the U.S. market. The Italian company ProGrade achieved commercial deployment in Europe in 2021 (CBJ, 2021), demonstrating feasibility, but carries no ANSI certification and has no stated U.S. timeline.

Our central argument is that the absence of a U.S. lead auto-belay reflects a *psychological equilibrium* sustained by three interlocking mechanisms. Climbers extend profound distrust to automated life-critical systems they cannot understand — a robust finding across the automation trust literature (Lee & See, 2004; Parasuraman & Riley, 1997; Hoff & Bashir, 2015). Gyms do not deploy technology their members will resist. Manufacturers do not invest in products whose markets are constrained by user psychology as much as by engineering cost. And the fear-of-falling research establishes that the climbers who most need to practice falls are exactly the ones

whose psychological barriers are highest and whose existing infrastructure provides the least support (Pijpers et al., 2006; Nieuwenhuys et al., 2008; Draper et al., 2010).

Figure 1 illustrates the self-reinforcing structure of this equilibrium.

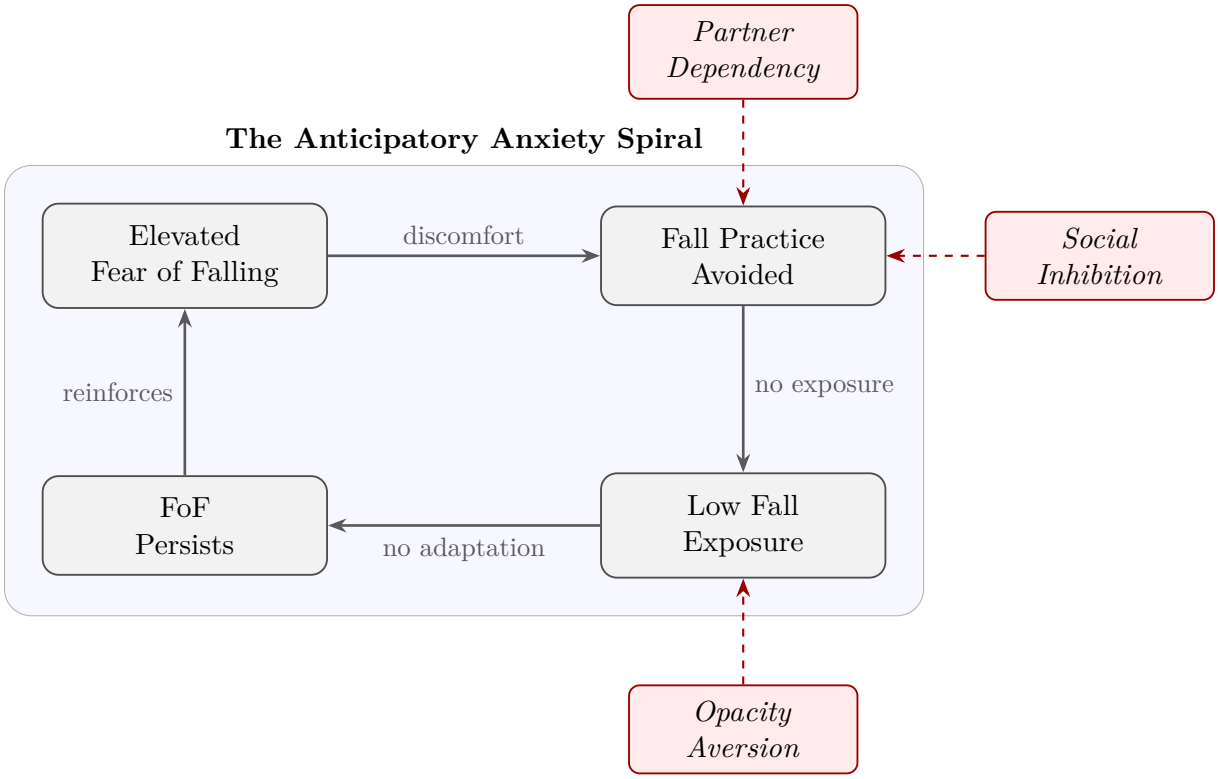

Figure 1: The anticipatory anxiety spiral in lead climbing. The four inner nodes form a self-reinforcing cycle: elevated fear of falling leads to avoidance of fall practice, which produces low fall exposure, which prevents fear adaptation and sustains elevated fear. Three structural barriers (red, dashed) feed into the cycle at distinct points: partner dependency and social inhibition restrict access to fall practice, while opacity aversion toward automated belay devices prevents the technological intervention most likely to break the cycle.

The paper proceeds as follows. Section 2 reviews the psychology and physiology of fear of falling in lead climbing. Section 3 examines algorithm aversion and opacity aversion as barriers to automated safety device adoption, and applies the Mayer et al. trust model to the lead auto-belay context. Section 4 analyzes the belay partnership as a social structure and documents belayer error as the primary failure mode in human-belayed climbing. Section 5 explains the technological gap and characterizes existing prior art including ProGrade. Section 6 introduces ARIA as a design response.

Section 7 discusses implications. Section 8 concludes.

2 Fear of Falling in Lead Climbing

2.1 The Physical Structure of Falling

In top-rope climbing, the rope runs from the climber's harness upward through an anchor at the top of the wall and back down to the belayer. A fall is arrested almost instantaneously: the climber descends perhaps 30–50 cm before the rope fully loads, limited primarily by dynamic rope stretch.

Lead climbing is categorically different. The climber ascends carrying the rope from below, clipping it to quickdraws bolted to the wall as they ascend. If they fall, they fall the entire distance above their last clipped protection *plus the same distance again below the clip*, giving a minimum fall distance equal to twice the height above the last protection. On a 10-meter indoor route with clips every 1.5 meters, a climber who falls before clipping the fifth bolt experiences approximately 3 meters of genuine freefall before the rope engages, followed by continued downward movement as rope stretch and belay-device payout absorb the load.

This is not a metaphorical distinction. It is a genuine freefall, physically and phenomenologically different from anything that occurs in top-rope climbing, and it is the defining anxiety-producing experience of lead climbing progression.

2.2 Fear of Falling as Distinct from Injury Risk

A foundational finding in the climbing psychology literature is that fear of falling (FoF) is not well-predicted by the actual probability or severity of injury (Pijpers et al., 2003; Brymer & Schweitzer, 2013; Immonen et al., 2018). On a well-maintained indoor wall with clean protection and an attentive belayer, a lead fall is objectively safe for the vast majority of climbers. Nonetheless, many climbers report significant resistance to falling that persists even after they have intellectually internalized the safety of the system.

Pijpers et al. (2003) demonstrated this in a foundational investigation using a processing efficiency theory framework: participants climbing identical routes at high height (6 m) versus low height (2 m) showed significantly increased anxiety, longer movement times, and altered visual search behavior at height, despite identical protection at both elevations. Pijpers et al. (2005) extended this to kinematics, showing height-induced fear altered movement fluency, limb velocity, and contact-force patterns on identical sequences. Pijpers et al. (2006) applied an ecological psychology framework, demonstrating that increased state anxiety at height changes perceived affordances such that routes that appear climbable at low height may no longer appear so at high height, even with identical physical holds. Nieuwenhuys et al. (2008) examined visual attention specifically, finding that higher perceived fall risk produced longer fixation times and slower movement despite identical holds and protection, with fine-coordination tasks degrading more than gross-strength tasks.

The conclusion is clear: a climber who intellectually knows the rope will catch them still experiences attentional, physiological, and motor disruption at their fear threshold. This is the normal operation of a pre-cognitive threat-response system that does not consult the rational mind before acting, and that cannot be countermanded by rational reassurance.

2.3 Physiological Evidence: Lead versus Top-Rope

The behavioral findings above are supported and extended by a body of psychophysiological research demonstrating that lead climbing, as distinct from top-rope, produces measurably elevated fear responses even when the physical route is identical.

Hodgson et al. (2009) measured plasma cortisol and state anxiety in climbers completing three conditions designed to elicit low, moderate, and high stress via different safety rope protocols analogous to top-rope versus lead configurations. Changes in cortisol and somatic anxiety increased linearly with stress level, establishing a tight relationship between rope protocol and physiological fear response.

Draper et al. (2010) conducted a direct within-subject comparison of on-sight lead versus top-rope on the same route in intermediate climbers, measuring heart rate, blood lactate, and self-reported anxiety. Lead climbing elicited higher peak and mean heart rate and higher anxiety than top-rope for the same physical task, supporting the claim that perceived fall consequence elevates psychophysiological stress beyond pure physical workload. A 2025 cross-sectional study (BMC Sports Science, 2025) replicated these findings in a larger sample, reporting significantly higher anxiety and physiological arousal during lead compared with top-rope, with climbing method as a primary determinant.

These findings have a direct design implication: physiological fear markers are objective, measurable outcomes against which ARIA-enabled training interventions could be evaluated in future empirical work.

2.4 The Efficacy of Fall Practice

The consistent answer across the literature to the question of what reduces FoF is controlled, progressive exposure to falling itself (Pijpers et al., 2003; Immonen et al., 2018). Across the Pijpers and colleagues research program, participants performing repeated trials at height showed gradual improvements in movement fluency and reductions in physiological anxiety markers within sessions, consistent with an exposure mechanism in which accumulated fall experience reduces the prediction error driving anticipatory anxiety.

Niedermeier et al. (2019) examined this prospectively in a structured fall-training program, measuring heart-rate variability (HRV) and salivary cortisol before and after graded fall exposure. Over the course of training, HRV patterns indicated reduced sympathetic dominance and salivary cortisol decreased, particularly in the group whose exposure progressed from lower to higher heights. A 2025 study comparing novice and advanced climbers (IJERPH, 2025) found that advanced climbers displayed lower anxiety and more favorable autonomic balance (HRV markers indicating better parasympathetic modulation), consistent with the interpretation that accumulated fall experience reshapes baseline psychophysiological fear responses.

Immonen et al. (2018) synthesize the broader action-sports literature using an ecological dynamics framework, arguing that repeated engagement with high-consequence environments is best understood as graded exposure and skill acquisition rather than risk tolerance, with the experienced climber having developed a more accurate and differentiated predictive model of the fall experience.

This framework establishes a clear design hypothesis: an effective tool for FoF reduction should enable high-frequency, low-friction fall practice in a controlled environment. A device that catches falls reliably without requiring a human partner removes the two primary access barriers to fall practice — partner availability and the social cost of falling repeatedly in front of another person — and should substantially accelerate fear adaptation.

2.5 The Anticipatory Anxiety Spiral

The barriers to fall practice do not operate independently. Their structural interdependence forms the self-reinforcing cycle illustrated in Figure 1. Elevated FoF makes fall practice uncomfortable. Partner dependency makes fall practice logistically constrained. Social inhibition makes intentional falling in front of a partner psychologically costly. Low fall-practice frequency prevents the physiological adaptation documented by Niedermeier et al. (2019) and Pijpers et al. (2003). Unadapted FoF keeps all three barriers in place.

This is why the absence of a lead auto-belay is psychologically significant rather than merely a technical inconvenience. Removing the partner-dependency constraint breaks the spiral at its most tractable point, directly enabling the high-frequency fall exposure the literature identifies as the primary fear-reduction mechanism.

3 Algorithm Aversion and the Opacity Problem

3.1 Trust in Safety-Critical Automation

The Technology Acceptance Model (TAM), proposed by Davis (1989) and extended by Venkatesh & Davis (2000), predicts technology adoption from perceived usefulness

and perceived ease of use. Extensions to safety-critical domains consistently identify a third factor: *perceived trustworthiness* under high-stakes failure conditions (Hoff & Bashir, 2015). For a device whose failure mode is a fall from height, perceived trustworthiness may be the dominant adoption variable, displacing usefulness and ease of use until a threshold of trust has been established.

Lee & See (2004) define trust in automation as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability," and identify three governing factors: *performance* (reliability history), *process* (transparency about how outputs are produced), and *purpose* (alignment between the automation's goals and the user's goals). Parasuraman & Riley (1997) identify disuse — under-reliance on automation not because of demonstrated failures but because of insufficient trust — as the primary risk in safety-critical adoption contexts. Hoff & Bashir (2015) confirm that perceived risk of failure is among the strongest moderators of automation trust, with higher perceived consequences substantially reducing willingness to rely on automated systems.

3.2 The Mayer Trust Model Applied to Lead Auto-Belay

Mayer et al. (1995) decompose perceived trustworthiness into three dimensions: *ability* (the trustee's competence to perform the relevant function), *benevolence* (alignment of the trustee's goals with the trustor's interests), and *integrity* (the trustee's adherence to acceptable principles). A separate dispositional factor — propensity to trust — reflects the trustor's general tendency to trust others and new systems prior to specific experience (Kołodziejczyk et al., 2021; Hancock et al., 2011).

This framework has been widely adopted in automation trust research (Kołodziejczyk et al., 2021; Lee & See, 2004) and maps directly onto the lead auto-belay adoption problem. Table 1 shows how specific ARIA design features target each dimension of perceived trustworthiness.

The cold-start trust problem — the challenge of building trust in a system that cannot accumulate a track record until it has been used — is directly addressed

Table 1: Mayer et al. (1995) trust dimensions mapped to ARIA design features

Dimension	**What it means for a belay device**	**ARIA design response**
Ability	The device reliably catches falls across all expected conditions	ANSI Z359.14-2021 compliance analysis; ratchet/pawl geometry sized for 8,000 N; 5.97× safety factor on tooth stress
Benevolence	The device's goals align with the climber's goal of returning safely to the ground	Independent mechanical catch with zero electronics dependency; two-factor confirmation prevents false positives; voice commands preserve climber agency
Integrity	The device behaves consistently and according to stated principles	Observable ratchet-and-pawl mechanism; inspectable by climbers; demonstrable without electrical power; behavioral continuity with human belay vocabulary
Propensity	The climber's prior disposition toward automated safety systems	Addressed by structured onboarding (Section 3.6) and mechanical transparency that permits trust formation before fall experience accumulates

by the ability and integrity columns of Table 1. Hancock et al. (2011) confirm in a meta-analysis of 29 studies that system performance, purpose, and process all significantly impact trust, particularly in early interactions. de Visser & Parasuraman (2011) show that structured interaction with clear feedback during early use helps maintain appropriately calibrated trust. For ARIA, this translates to a specific design requirement: the mechanical transparency of the catch mechanism must allow a climber to build an *integrity*-based mental model before they have accumulated any performance-based experience.

3.3 Algorithm Aversion

Dietvorst et al. (2015) demonstrated algorithm aversion across multiple forecasting experiments: participants abandoned accurate algorithms after observing a single error, switching to human judgment even when the human made the same type of error more frequently. Two mechanisms drive this: an accountability mismatch (human errors are forgiven in ways algorithmic errors are not) and categorical unreliability (an algorithm that errs is mentally recategorized as broken).

Dietvorst et al. (2016) provide the most practically important finding: algorithm aversion is substantially reduced when users are given even minimal control over the algorithm's outputs. The ability to adjust, modify, or override the system's decisions dramatically increases willingness to rely on it, even when the underlying reliability is unchanged. Kim & Sundar (2022) confirm that in high-stakes domains, algorithm aversion dominates over automation bias: a single observed failure produces lasting reluctance even after the failure has been explained and corrected.

For a lead auto-belay, these findings have concrete implications. A single documented incident involving any automated belay device is likely to produce disproportionate and persistent distrust across the entire product category, consistent with the documented industry response to top-rope auto-belay incidents (Merritt & Ilgen, 2008; Climbing Wall Association, 2023). Dietvorst et al. (2016)'s finding motivates ARIA's voice command system directly: by giving climbers real-time control over rope management, the device reframes the climber's relationship from passive recipient of algorithmic

decisions to active director of a responsive tool.

3.4 Opacity Aversion

Dzindolet et al. (2002) demonstrated that automation trust is significantly higher when users can observe the automation's decision process, even when that observation provides no additional information about reliability. Wortham et al. (2017) argue that observability of an automated system's behavior is a primary determinant of whether users develop appropriate trust. Baltrusaitis et al. (2022) synthesize evidence showing that in safety-critical domains where failure consequences are severe, opacity tends toward disuse: users avoid what they cannot understand.

A human belayer is perceptually transparent: the climber watches the rope being managed, reads the belayer's attention level, and forms a real-time trust judgment. An automated device is a black box. The climber's pre-cognitive threat-assessment system cannot be satisfied by technical certification data. That information exists at the rational-cognitive level. The visceral trust required to commit to a difficult move above the last clip requires a qualitatively different kind of evidence: direct perceptual experience of the system working, accumulated over time.

3.5 The Novelty Penalty

New safety devices systematically face higher skepticism than their eventual reliability warrants because users lack the embodied experience needed to build trust (McCammon, 2004; Haslam & Stubbs, 2006; Zoss et al., 2015). For a lead auto-belay, the novelty penalty is compounded: the first time a climber clips in and steps off the wall, they simultaneously manage FoF, uncertainty about device reliability, and uncertainty about the quality of their own judgment of the device. Three simultaneous cognitive loads, activated at maximum physical and psychological demand. The design response is not to wait for cultural familiarity to accumulate, but to minimize each load through deliberate architectural choices.

3.6 Structured Onboarding as Trust Calibration

de Visser & Parasuraman (2011) show that structured interaction and clear feedback during early use help maintain calibrated trust in imperfect automation. Abe & Richardson (2020) demonstrate in a drone simulator that explicit trust calibration cues injected during early sessions significantly shift behavior compared with static information, indicating that interactive onboarding accelerates appropriate trust formation. Chavaillaz et al. (2016) show that training with clear information about system limitations improves trust calibration particularly under cognitive load — precisely the conditions of a first lead climb on an unfamiliar device.

These findings motivate a specific onboarding protocol for ARIA, which is described in Section 6.4: a graded sequence of low-stakes exposures that allows trust to accumulate from the ability and integrity dimensions before any fall experience is required.

4 The Belay Partnership as a Social Structure

4.1 Beyond Catching: The Psychological Functions of the Belayer

The human belayer serves functions that extend well beyond arresting falls.

Attention and witness. The belayer watches the climber throughout the ascent. Being attended to by a focused partner communicates to the climber's pre-cognitive social-monitoring system that they are not alone in a vulnerable situation. Seifert et al. (2022) found through qualitative interviews with elite climbers that a positive, low-tension relational atmosphere with the belayer is a prerequisite for optimal performance, and that relationship tensions carry over directly into movement quality and decision-making.

Communication and preserved agency. The belay partnership is interactive and responsive. A climber calls "take," and the belayer responds. This responsiveness preserves the climber's sense of agency over their own safety — the rope is doing something *for* them at their direction, not *to* them on the system's terms. Celsi et al. (1993) and Brymer & Schweitzer (2013) document that expert action-sport

athletes consistently identify perceived control and preparation as the psychological mechanisms through which they maintain functional rather than paralyzing fear responses.

Reciprocity and implicit costs. The belay relationship is typically reciprocal. A climber who wants to work a hard route forty times must reckon with whether they are imposing excessively on their partner. This resource constraint shapes training decisions in ways that are rarely explicit but consistently real.

Trust through relationship. The belayer is typically a known individual with an established relationship. Seifert et al. (2022) document that climbers organize training around compatible partners and that interpersonal fit is a practical constraint on training volume. Replacing a person with a machine does not merely replace a technical function — it replaces a relationship, and the trust accumulated through shared experience must be rebuilt from scratch with the device.

4.2 Belayer Error as the Primary Failure Mode

A critical but underappreciated asymmetry in the trust calculus is that the human belay system is itself a significant source of risk. Schöffl et al. (2024) analyzed 9,885 climbing incidents from German Alpine Association records, finding that in indoor wall climbing, belayer error is more frequently reported as the cause of injury than in outdoor climbing, and that in top-rope climbing specifically, longer falls occur primarily in the case of a belayer's error rather than equipment failure.

The Climbing Wall Association's analysis of indoor climbing incidents identifies distraction, socializing, and autopilot belaying as recognized root causes of accidents, recommending system interventions such as belay gates and structured orientations to interrupt inattentional patterns (Climbing Wall Association / Heshka, n.d.). This finding is theoretically grounded in the automation literature: Parasuraman & Riley (1997) document that operators monitoring a largely reliable process experience vigilance decrements and inattentional blindness, a pattern that maps directly onto a belayer monitoring routine falls over the course of a gym session.

Industry data indicate that when correctly clipped and maintained, automated belay devices fail extremely rarely, and that the majority of auto-belay incidents are attributable to user error at the clip-in step rather than mechanical device failure (Climbing Wall Association, 2023). The Norwegian Climbing Federation's analysis of national incident reports documented a striking decrease in auto-belay accidents following new guidelines, with only seven such incidents in 2024 and three occurring before guideline publication (Norwegian Climbing Federation, 2025).

This asymmetry — human belayers introducing continuous, ongoing error risk throughout the climb; automated devices introducing primarily a discrete, preventable clip-in error — is not reflected in climbers' subjective trust assessments, consistent with the algorithm aversion literature's finding that human errors are forgiven more readily than algorithmic ones (Dietvorst et al., 2015; Kim & Sundar, 2022).

4.3 The Partner Availability Constraint

The practical consequence of the human-partner requirement is a hard structural constraint on training frequency. The Climbing Wall Association Indoor Climber Survey (2019) indicated that a substantial fraction of indoor climbers prefer bouldering partly for its lack of partner requirements, providing indirect evidence that partner dependence is a meaningful barrier to participation in roped climbing. Gym staff and managers consulted during the development of ARIA consistently described lead walls running at substantially lower utilization than their physical capacity would support, with bouldering areas maintaining high utilization throughout operating hours and lead areas spiking during organized classes and declining during open gym. The wall is not the constraint. The human is the constraint.

4.4 Social Inhibition of Fall Practice

Beyond limiting access frequency, the belayer's social presence may directly inhibit the fall practice that reduces fear. Zajonc (1965) established that observer presence improves performance on well-learned tasks but impairs performance on novel or anxiety-provoking ones. Bond & Titus (1983) confirmed this across a meta-analysis of

241 studies. For a climber specifically trying to practice falling — an uncomfortable activity that visibly marks them as struggling — an observing partner is more likely to impair than to facilitate.

Seifert et al. (2022) report that elite climbers consistently describe performing best when the social dimension of the belay partnership is relaxed and non-evaluative. The inverse is also evident: relationship stress with a belayer directly impairs climbing execution and risk-taking. For the intermediate climber attempting deliberate fall practice, the evaluative presence of a more experienced partner represents exactly the social pressure most likely to suppress the behavior the climber most needs to perform.

An automated device that removes the observing partner while preserving the catching function may therefore be more conducive to fall practice than the human belay relationship it replaces — not despite its impersonality, but in part precisely because of it.

5 The Technologically Maintained Gap

5.1 The Prior-Art Landscape

Three prior attempts define the lead auto-belay landscape.

Auroco (Germany, 2016). Auroco announced a portable lead auto-belay to significant trade-press attention, with *Climbing Magazine* labeling it a "game changer" (*Climbing*, 2016). The product never reached commercial release, with no technical documentation explaining the failure publicly available.

Lead Solo (academic prototype, c. 2020). A mechanical engineering student project adapted a centrifugal-clutch fall-arrest mechanism to a lead rope spool. The author communicated directly with the designer in early 2026. The catch mechanism functioned when rope tension was within calibrated range, but the tensioning subsystem prevented consistent engagement and the project never progressed to dynamic fall testing. Beyond the calibration problem, the centrifugal-clutch architecture has a fundamental incompatibility with ANSI Z359.14: brake force drops as ω^2 as the

spool decelerates during arrest, causing arrest distance to approach infinity — a structural incompatibility that eliminates an entire class of design approaches from ANSI-compliant consideration.

ProGrade (Italy, 2021–present). ProGrade, founded by three Italian engineers in 2017, is the only device to achieve commercial deployment (CBJ, 2021; *Climbing*, 2021). The device launched at UrbanWall gym in Milan in September 2021, recording approximately 700 uses in its first week and 3,700 uses over two months (GearJunkie, 2021). User reception was broadly positive; the company reported that even initially skeptical climbers were convinced and satisfied after using the device (CBJ, 2021). ProGrade is CE certified and has been tested to UIAA regulations by the Italian certification body Dolomiticert (*Climbing*, 2021). As of 2026, the device has expanded to several European gyms.

ProGrade's achievement is genuinely important: it established that commercial deployment is mechanically feasible and that real climbing gym populations will adopt a lead auto-belay. The company's data directly contradict any assumption that climbers will categorically reject the technology — the adoption problem is real but not insurmountable.

However, ProGrade's architecture involves specific limitations that the ARIA design directly addresses:

(i) Software-dependent catch. ProGrade's locking mechanism operates by rope friction controlled entirely by software and motors (GearJunkie, 2021; ProGrade, 2022). There is no independent mechanical layer requiring zero electronics. ARIA addresses this with a physically independent ratchet-and-pawl Layer 1 that operates without electrical power.

(ii) No U.S. certification pathway. CE certification satisfies EU requirements but not ANSI Z359.14-2021. ProGrade has stated no timetable for U.S. market entry (GearJunkie, 2021).

(iii) No climber agency interface. ProGrade manages the rope autonomously;

the climber cannot issue commands. There is no voice command system, no mechanism to call "take" or "slack," and no computer vision anticipation of clipping. As established in Sections 4.1 and 3.2, preserved agency directly addresses the benevolence dimension of perceived trustworthiness.

(iv) Reactive slack management. ProGrade's rope-movement sensor reacts after the rope has already moved; it cannot anticipate an imminent clip. Industry reviewers noted the device appeared to maintain tighter tension than a skilled human belayer (GearJunkie, 2021). ARIA's computer vision proactively pre-feeds slack when a clipping gesture is detected.

Table 2: Lead auto-belay prior art: architectural comparison

Feature	**Auroco**	**Lead Solo**	**ProGrade**	**ARIA**
Commercially deployed	No	No	Yes (EU)	No (POC)
U.S./ANSI certification	No	No	No	Designed for
Independent mech. catch	Unknown	Yes*	No	Yes
Voice commands	No	No	No	Yes
Computer vision	No	No	No	Yes
Climber agency interface	No	No	No	Yes
Two-factor confirmation	No	No	No	Yes

*Centrifugal clutch; ANSI Z359.14 incompatible due to the ω^2 brake-force relationship during arrest deceleration.

5.2 Why No U.S. Lead Auto-Belay Exists

ProGrade's existence clarifies the precise nature of the U.S. gap. The problem is not mechanical impossibility. The U.S. gap reflects: (1) the gap between CE/UIAA and ANSI Z359.14-2021; (2) a product liability environment rendered more conservative by the 2023 verdict against a top-rope auto-belay manufacturer; and (3) the absence

of a device architecture that directly addresses the psychological barriers identified in Sections 3 and 4 as constraints on adoption even after regulatory hurdles are resolved.

5.3 The Psychological Equilibrium

The gap is stable not because of engineering cost alone. The psychological adoption barriers constitute a second-order deterrent: even a certified device would face the documented distrust of climbers toward opaque life-critical automation (Lee & See, 2004; Dzindolet et al., 2002), risk aversion from gym operators, and liability concerns from insurers. These factors form the chicken-and-egg structure shown in Figure 1: the device cannot gain trust without widespread use, and cannot gain widespread use without demonstrated trust. Breaking this equilibrium requires a sufficiently compelling small-scale demonstration — the approach taken by ARIA's proof-of-concept program.

6 ARIA: A Design Response to Psychological Constraints

ARIA (Autonomous Rope Intelligence Architecture) is a hybrid mechanical–electronic lead auto-belay developed as a direct response to the psychological and technical constraints described above. Complete engineering justification and ANSI compliance analysis are documented in Kofman (2026a).

6.1 Design Principles Derived from the Literature

Four principles were extracted directly from the psychological findings:

1. **Mechanical transparency** (addressing opacity aversion (Dzindolet et al., 2002; Wortham et al., 2017) and the *integrity* dimension of the Mayer model (Mayer et al., 1995)).

2. **Preserved agency** (addressing the *benevolence* dimension (Mayer et al., 1995) and the action-sport agency literature (Celsi et al., 1993; Brymer & Schweitzer, 2013)).

3. **Layered and legible safety** (addressing the *ability* dimension (Mayer et al., 1995) and disuse risk (Parasuraman & Riley, 1997)).

4. **Behavioral continuity** (reducing the novelty penalty and leveraging transferred trust (Merritt & Ilgen, 2008; Dietvorst et al., 2016)).

6.2 Three-Layer Architecture

ARIA is organized around three strictly independent functional layers. Failure in any outer layer cannot compromise inner layers.

Layer 1 — Mechanical catch (no electronics required). An inertia-triggered ratchet-and-pawl mechanism arrests falls through purely mechanical means. The flyweight assembly detects rope acceleration above the fall-detection threshold; the blocker bar retracts and the pawl engages the ratchet wheel within one tooth pitch (26 mm). No electrical power is required. Engineering analysis confirms ANSI Z359.14-2021 compliance: peak arrest force 5,360 N against the 8,000 N limit; arrest distance 54 mm against the 813 mm limit; peak deceleration 5.48 g against the 10 g limit (Kofman, 2026a).

Layer 2 — STM32F411 safety layer. An STM32F411 microcontroller manages the BLDC motor, a 50 kg load cell via HX711 amplifier, an AS5048A magnetic encoder, and the eight-state ARIA state machine. All suggested state transitions from the intelligence layer are validated against live sensor data before execution. Safety-critical transitions require two-factor confirmation: a voice command followed by load-cell confirmation (>200 N within 500 ms).

Layer 3 — ESP32-S3 intelligence layer. A XIAO ESP32-S3 Sense manages voice command detection (Edge Impulse wake-word model, ≥85% confidence) and computer vision clipping detection (OV2640 camera). The command vocabulary — `TAKE`, `SLACK`, `LOWER`, `REST`, `WATCH ME`, `UP`, `CLIMBING` — mirrors verbatim the vocabulary of conventional human belaying.

6.3 How the Architecture Responds to Psychological Constraints

Opacity aversion. The ratchet-and-pawl mechanism is physically inspectable by gym staff and demonstrable to climbers while unpowered. The catch operation can be explained in one sentence and directly observed through an inspection window. A gym operator introducing ARIA to a skeptical member can show the ratchet wheel, manually demonstrate the pawl engaging, and allow the member to feel the engagement happen. This builds *integrity*-based trust before any fall experience is accumulated (Dzindolet et al., 2002; Mayer et al., 1995).

Algorithm aversion and two-factor confirmation. The most psychologically dangerous failure mode is a false positive. Two-factor confirmation substantially reduces this risk and makes the governing rule communicable: "the rope only locks if you say `take` *and* physically weight it." Dietvorst et al. (2016) demonstrate that user control over an algorithm's outputs substantially reduces algorithm aversion; two-factor confirmation makes the climber an active participant in the safety-critical transition.

Preserved agency. The voice command system directly addresses the *benevolence* dimension of the Mayer model by demonstrating that ARIA's goals are aligned with the climber's — the device responds to what the climber wants, not to what the algorithm has decided.

Behavioral continuity. The command vocabulary is borrowed verbatim from human belaying. The experience of hanging on the rope is physically identical to hanging on a human belay. ARIA does not create a new user experience; it transparently replaces one function within a familiar one, allowing existing trust toward the belayer relationship to partially transfer (Merritt & Ilgen, 2008).

6.4 Proposed Onboarding Protocol

Drawing on the structured onboarding literature (de Visser & Parasuraman, 2011; Abe & Richardson, 2020; Chavaillaz et al., 2016) and the graded exposure literature (Pijpers et al., 2003; Niedermeier et al., 2019), the ARIA onboarding protocol follows four stages:

Stage 1. Inspection and demonstration. Gym staff demonstrate the ratchet-and-pawl mechanism physically, with the device unpowered, before the climber clips in. This establishes *integrity*-based trust through direct perceptual experience of the mechanism.

Stage 2. Static hang. Climber clips in, weights the rope at ground level, and calls `take`. They hang for 30 seconds, experiencing the two-factor confirmation and the device's mechanical feel without height exposure.

Stage 3. Low-route falls. Climber ascends to the first clip only and takes two to three intentional falls. Device behavior under dynamic load is experienced at minimum height, building performance-based trust before anxiety levels escalate.

Stage 4. Progressive height. Climber progresses to higher clips, taking at least one intentional fall per session. This mirrors the graded fall-practice curricula that the exposure literature identifies as most effective for FoF reduction (Pijpers et al., 2003; Immonen et al., 2018; Niedermeier et al., 2019).

This protocol is not merely a safety procedure; it is a trust calibration sequence grounded in both the automation trust literature and the fear-of-falling literature.

6.5 Hardware Summary

7 Discussion

7.1 Implications for Safety-Critical Automation Design

Three generalizable lessons emerge from the ARIA case study.

Adoption is not primarily determined by reliability. The opacity-aversion and algorithm-aversion findings (Dzindolet et al., 2002; Dietvorst et al., 2015) establish that adoption of life-critical automation requires parallel investments in *legibility* — making the system's mechanism perceptible to users — independent of investments in reliability.

Table 3: ARIA primary hardware components

Component	**Part**	**Function**
Safety MCU	STM32F411 Black Pill	State machine, motor FOC, sensor fusion
Intelligence MCU	XIAO ESP32-S3 Sense	Voice detection, computer vision, BLE
Motor	T-Motor GB54-2 BLDC	Active slack management
Gearbox	30:1 planetary	Torque multiplication
Load cell amp.	HX711 + 50 kg cell	Real-time rope tension
Encoder	AS5048A magnetic	Spool position (<1 mm resolution)
Motor controller	VESC Mini 6.7	Field-oriented motor drive

Table 4: ARIA ANSI Z359.14-2021 compliance summary (100 kg test mass)

Parameter	**ANSI Limit**	**ARIA Calculated**	**S.F.**
Peak arrest force	≤8,000 N	5,360 N	1.49×
Average arrest force	≤6,000 N	2,388 N	2.51×
Maximum arrest distance	≤813 mm	54 mm	15.1×
Peak deceleration	≤10 g	5.48 g	1.82×
Ratchet tooth stress	material S.F.	301 MPa	5.97×
Main shaft (von Mises)	≤1,800 MPa	835 MPa	2.16×

A perfectly reliable system that users cannot mentally model will be systematically under-adopted.

The Mayer dimensions provide a design roadmap. Decomposing trust into ability, benevolence, and integrity (Mayer et al., 1995) provides concrete and actionable design targets. Ability is addressed by certification and reliability data. Benevolence is addressed by preserved agency and fail-safe architecture. Integrity is addressed by mechanical transparency and behavioral continuity. Each dimension can be targeted by specific design choices, and each has a different evidentiary basis that reaches a different cognitive system in the user.

Social function is part of the safety function. A lead auto-belay that catches falls but strips out the social dimensions of the belay relationship without providing functional substitutes will face adoption resistance that reliability data cannot overcome (Seifert et al., 2022).

7.2 Implications for Climbing Pedagogy

The fear-of-falling literature (Pijpers et al., 2003; Nieuwenhuys et al., 2008; Draper et al., 2010; Niedermeier et al., 2019) establishes that progressive fall practice is the most effective FoF intervention. A lead auto-belay changes the economics of fall practice fundamentally: with an automated device, a climber can take dozens of intentional falls at no social cost and at any time the gym is open. This changes accessible training volume for fear adaptation by an order of magnitude — if the device achieves adoption.

The physiological markers established in Section 2.3 (cortisol, HRV, heart rate) provide objective outcome variables for future longitudinal studies comparing FoF adaptation trajectories in human-belayed versus auto-belayed conditions, filling a significant gap in both the sport psychology and human factors literatures.

7.3 Limitations

Several limitations should be acknowledged. The empirical literature on FoF specifically in indoor lead climbing is relatively sparse; the foundational studies used experimental heights (2–6 m) lower than typical gym lead routes (10–15 m), and generalizability has not been directly tested. The trust and algorithm aversion analyses draw on studies from informationally analogous but not sensorially analogous contexts. ProGrade's user acceptance data are self-reported marketing data, not peer-reviewed epidemiology. ARIA remains in the pre-hardware proof-of-concept phase, and all engineering analysis is theoretical pending physical validation.

Future work should prioritize: (1) survey studies measuring opacity aversion and algorithm aversion toward lead auto-belays as distinct constructs in gym populations; (2) longitudinal trust development studies in actual deployed device users; (3) controlled comparisons of fall-practice frequency and FoF reduction in human-belay versus auto-belay conditions, using the physiological markers established in Section 2.3; (4) epidemiological comparison of incident rates between human-belayed and auto-belayed lead climbing.

8 Conclusion

Indoor lead climbing is the most technically demanding and progression-relevant discipline available on a gym wall, and the one most resistant to the technological evolution that has made every other gym activity accessible to solo visitors. This paper has argued that the absence of a U.S. lead auto-belay reflects a psychological equilibrium maintained by three interlocking forces: pre-cognitive fear that resists rational reassurance; documented distrust of opaque automated systems in life-critical contexts; and the social and psychological functions of the human belay partnership that any automated replacement must take seriously.

The Mayer et al. trust model provides a concrete roadmap for addressing this equilibrium at the design stage. Ability, benevolence, and integrity each require distinct design responses, and each reaches a different cognitive system in the user. No amount

of reliability data addresses the integrity deficit created by an opaque mechanism. No engineering sophistication substitutes for the agency preserved by a responsive voice interface. And no certification eliminates the cold-start trust problem — that challenge can only be addressed by structured onboarding that creates the embodied experience needed to build trust before it is demanded.

ProGrade demonstrated in 2021 that the technology acceptance problem is real but surmountable: real climbers, including initially skeptical ones, will adopt a lead autobelay when given access to one. What remains is to build a device that addresses the full psychological architecture of the problem, not merely its mechanical core.

Trust in a life-critical system is not a property of the device. It is a property of the relationship between the device and the person whose life depends on it. Engineering that relationship — not merely the mechanism — is the work that makes autonomous safety systems worth building.

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A ARIA Hardware and Architecture Summary

Tables 3 and 4 appear in the main text.

Eight-state machine: `IDLE` → `CLIMBING` → `CLIPPING` → `TAKE` → `REST` → `LOWER` → `WATCH_ME` → `UP`. Safety-critical transitions require two-factor confirmation: voice command at ≥85% model confidence *and* load cell >200 N within a 500 ms window.

Repository: github.com/jonathan-kofman/aria-auto-belay